

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE – PILANI, HYDERABAD
CAMPUS
EEE/INSTR F342: POWER ELECTRONICS, SEM-II, 2025-26

TUTORIAL- 1

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Diode Characteristics and Single-phase half wave diode rectifier

- Q1. A PN junction silicon diode has a reverse saturation current rating of 50nA at 32°C . What should be the value of the forward anode current for a forward voltage drop of 0.5V ? Find the thermal voltage at 32°C ?
- Q2. If the reverse recovery storage charge for a power diode is $95\mu\text{C}$ and the peak reverse recovery current is 50A , then calculate the rate of change of current. Also identify whether the diode is hard recovery or soft recovery.
- Q3. Derive the Fourier expression for the input current waveform of a half wave rectifier.
- Q4. Derive the Fourier expression for the output voltage waveform of a full wave rectifier.
- Q5. A single-phase, 230V , 1KW lamp load is connected across single-phase 230V , 50Hz supply through a power diode. Calculate the power delivered to the lamp load. Also find a) the peak diode current b) rectification efficiency c) RMS V_{AC} d) Form Factor e) Ripple Factor f) Input Power Factor (assuming ideal condition for the diode)
- Q6. A half-wave rectifier circuit with resistive load of 750Ω and a diode with an internal resistance of 15Ω is supplied from a 110V RMS single-phase AC supply. Find (a) Peak load current I_m
(b) DC load current I_{DC}
(c) AC Load current I_{RMS}
(d) DC Load voltage E_{DC}
(e) Total input power to the circuit P_{AC}
(f) Ripple Factor
(g) Percentage regulation from no load to the given load (%R)